1. Electric Charge & Electric Current

* Charge Q = ne (e = 1.6 x 10^-19 C).
* Current I = Q/t, SI unit: Ampere (A).
* Direction of current is opposite to flow of electrons.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Ammeter measures current, connected in series, low resistance.

> 1 Ampere = 1 Coulomb per second.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

2. Electric Potential & Potential Difference

* Work done in moving unit positive charge: $V = W/Q$.
* SI unit of potential difference: Volt (V).
* Voltmeter measures V, connected in parallel, high resistance.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> 1 Volt = 1 Joule / 1 Coulomb.

> Cell/battery maintains potential difference across circuit.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

3. Ohm Law & V-I Characteristics

* Statement: V = IR at constant temperature.
* Resistance R = V/I, SI unit: Ohm (Omega).
* V-I graph for ohmic conductor is a straight line passing through origin.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Slope of V-I graph gives Resistance R.

> 1 Ohm = 1 Volt / 1 Ampere.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

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Q4. Solve numerical questions based on standard Board exam paper patterns.

4. Factors Affecting Resistance & Resistivity

* Resistance proportional to length (R ~ l).
* Resistance inversely proportional to area (R ~ 1/A).
* Resistivity rho = RA/l, SI unit: Ohm-meter (Omega*m).

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Resistivity depends only on material & temperature.

> Alloys have higher resistivity than pure metals (used in heating coils).

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

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5. Resistors in Series Combination

* Same current I flows through each resistor.
* Total Potential Difference V = V1 + V2 + V3.
* Equivalent Resistance Rs = R1 + R2 + R3.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Equivalent resistance in series is larger than any individual resistance.

> If one appliance breaks, whole circuit breaks.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

6. Resistors in Parallel Combination

* Same potential difference V across each resistor.
* Total Current $I = I_1 + I_2 + I_3$.
* $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Equivalent resistance in parallel is smaller than smallest individual R.

> Used in household wiring so each appliance operates independently.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

7. Heating Effect of Electric Current

* Work done $W = VIt = I^2 R t$.
* Joule Law of Heating: $H = I^2 R t$.
* Heat produced proportional to I^2, R, t.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Applications: Electric Iron, Heater, Toaster, Fuse wire.

> Fuse wire has low melting point & high resistance.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

8. Electric Power & Commercial Unit of Energy

* Electric Power $P = W/t = VI = I^2 R = V^2 / R$.
* SI unit of Power: Watt (W) = 1 Joule/second.
* Commercial Unit: 1 kilowatt-hour (1 kWh) = 3.6×10^6 Joules.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> 1 kWh is called 1 Unit of electricity.

> Power rating (e.g. 100W, 220V) indicates energy consumption rate.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

Q1. Explain the main concept with suitable diagram and units.

Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

9. Important Circuit Diagrams & Numericals

* Series vs Parallel connection circuit schematics.
* Calculating total current & power in complex circuits.
* Solving board exam numerical problems step-by-step.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Always convert time to seconds in $H = I^2 Rt$ formula.

> Power $P = V^2 / R$ when appliances are in parallel across same V.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

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Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

Q4. Solve numerical questions based on standard Board exam paper patterns.

10. One Page Quick Revision & Exam Booster

* All key formulas: Q=ne, I=Q/t, V=W/Q, V=IR, rho=RA/l, P=VI, H=I^2Rt.
* Definitions, SI units, and conversion factors.
* Previous 5 Years Board Exam Questions (BSEB Class 10).

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

> Series vs Parallel comparative table.

> Memory tricks for Ohm Law and Power formulas.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

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Q2. Write short notes on key phenomena and practical applications.

Q3. Differentiate between primary principles and real-world examples.

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